



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 12

Assignment: Integrated Object-Oriented Modelling and Design

PROGRAMME

BSc Information Communication and Technology

COURSE CODE / YEAR

BICT 3202 · Year 3

LECTURER

Francis Fweta — ICT Lecturer

DELIVERY

2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 12 of 16



Contents

ASSIGNMENT INSTRUCTIONS	3
Section A — Short-Answer Questions (20 marks)	3
Section B — Application Questions (20 marks)	3
Section C — Case Study (10 marks)	4
Marking Scheme	4
Submission & Academic Integrity	4

ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 12 module and your lecture notes before attempting each question.
- Where you draw UML diagrams, use correct notation — actors, multiplicities, states, messages.
- Show how your models are consistent: every class in a sequence diagram must exist in the class model.
- Cite the OMG UML 2.5.1 specification where relevant.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define integrated object-oriented modelling. [2 marks]

Question A2

Explain why the different UML models of a system must be consistent. [3 marks]

Question A3

State the main question answered by each of the use-case, class, sequence, activity and state models. [3 marks]

Question A4

Define model consistency and give one example. [2 marks]

Question A5

Explain traceability and why it is valuable in integrated modelling. [2 marks]

Question A6

What is a design pattern and how does it support reuse? [2 marks]

Question A7

Differentiate between a class diagram and an object diagram. [2 marks]

Question A8

Explain how sequence diagrams and class diagrams complement each other. [2 marks]

Section B — Application Questions (20 marks)

Question B1 (12 marks)

Consider a simplified student registration system with Student, Course and Registration.

- (a) Draw a class diagram showing classes, attributes and operations. [4 marks]
- (b) Draw a sequence diagram for 'register for course'. [4 marks]
- (c) Draw a state diagram for Registration. [4 marks]

Question B2 (8 marks)

Consider the same registration system.

- (a) Explain how the three models are consistent. [4 marks]
- (b) Explain how they tell different parts of the story. [4 marks]

Section C — Case Study (10 marks)

Consider a complete university management system with all models from Weeks 1-12.

As an object-oriented analyst:

- (a) Explain how all models fit together to describe the complete system. [4 marks]
- (b) Identify potential inconsistencies and how to resolve them. [3 marks]
- (c) Explain the value of integrated modelling for understanding and maintaining the system. [3 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (8 questions)	20
B	Application questions (12 + 8)	20
C	Case study — complete system	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.
- Ensure your diagrams are consistent with each other.
- Cite any sources you consult using the format used in the Week 12 module references.